

HANNAH ARNOLD

First Class BSc (Hons) Artificial Intelligence and Robotics Graduate

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PROFESSIONAL PROFILE

First Class Artificial Intelligence and Robotics graduate with a strong interest in embedded software engineering, robotics systems and low-level programming. Through academic projects and independent technical development, I have gained practical experience designing and programming embedded systems using STM32 microcontrollers, integrating sensors and actuators, and troubleshooting both hardware and firmware issues.

My technical interests include embedded C/C++ development, hardware-software integration, robotics, automation and intelligent systems. I am particularly motivated by solving complex engineering challenges and developing reliable software solutions that interact directly with physical hardware. I am now seeking a graduate role within embedded software, robotics, electronics or software engineering where I can continue to develop my technical expertise while contributing to innovative projects.

TECHNICAL CAPABILITY

Embedded Systems & Electronics	Practical experience developing embedded applications using STM32 microcontrollers, incorporating GPIO, timers and peripheral interfaces. Knowledge of sensor and actuator integration, I2C communication, breadboard prototyping, external power supplies and hardware-firmware fault finding.
Software Development	Proficient in C, C++ and Python, with experience applying structured programming principles, algorithm design and low-level software development. Familiar with Git, GitHub, Visual Studio and technical problem-solving within project-based environments.
Professional Strengths	Confident communicator with strong teamwork, organisation and leadership skills developed through university projects, employment and sports captaincy. Adaptable, reliable and able to maintain accuracy and professionalism under pressure.

EDUCATION

Staffordshire University | September 2023 - May 2026

BSc (Hons) Artificial Intelligence and Robotics - First Class Honours

Relevant academic work included robotic programming, artificial intelligence and chatbot development, embedded systems design and software development using Python, C and C++. Practical assignments involved STM32 microcontrollers, GPIO, timers, sensors, I2C communication protocols and the integration of hardware and software components.

City of Stoke-on-Trent Sixth Form College | September 2021 - July 2023

A-Level studies included Mathematics, Physics, Further Mathematics and Extended Project Qualification (EPQ).

TECHNICAL PROJECTS

Final Year Project: Sensor-Based Embedded System for Roller Coaster Ride Path Analysis | October 2025 - May 2026

Designed and developed an embedded system investigating how roller coaster experiences could be personalised without relying on virtual reality. The system used multiple sensors and a camera module to collect rider and environmental data, supporting the selection of a suitable ride path based on user response.

The project involved embedded software development in C for STM32 microcontrollers, hardware-software integration, sensor interfacing, electronic circuit testing and diagnosis of hardware and firmware issues. This strengthened my understanding of real-world embedded systems, practical electronics and software reliability.

TECHNICAL PROJECTS CONTINUED

Personal Technical Portfolio | May 2026 - Present

Portfolio: xxhansterxx.github.io

Developing a personal technical portfolio to document ongoing learning and showcase projects across software engineering, embedded systems and web development. The site demonstrates independent study, technical curiosity and continuing development beyond university coursework.

EMPLOYMENT EXPERIENCE

Retail Assistant, Primark | April 2022 - Present

Working in a fast-paced retail environment has developed my customer service, communication and teamwork skills. I am trusted to work across multiple departments and operate till systems accurately, while adapting to changing priorities during busy trading periods.

This role has strengthened my reliability, attention to detail and ability to remain composed under pressure. These transferable skills support my transition into a professional technical environment where collaboration, accountability and consistent performance are essential.

LEADERSHIP AND WIDER EXPERIENCE

Committee Member and Captain, Macclesfield Warriors Hockey Club | December 2024 - Present

As Team Captain and Committee Member, I contribute to team organisation, communication and fixture coordination. The role requires calm decision-making, leadership under pressure and the ability to support others while maintaining a positive and professional team environment.

Engineering Society, Staffordshire University | September 2023 - May 2024

Participated in practical engineering workshops involving electronics assembly, soldering and prototype development. Activities also included CAD-supported 3D-printing and laser-cutting projects, supporting broader practical engineering awareness.

SELECTED TECHNICAL KNOWLEDGE

Languages	Embedded & Electronics	Tools	Methods
C, C++, Python, HTML	STM32, GPIO, Timers, I2C, Sensors, Actuators	Git, GitHub, Visual Studio, Portfolio website	Structured programming, Algorithm design, Debugging, Circuit testing